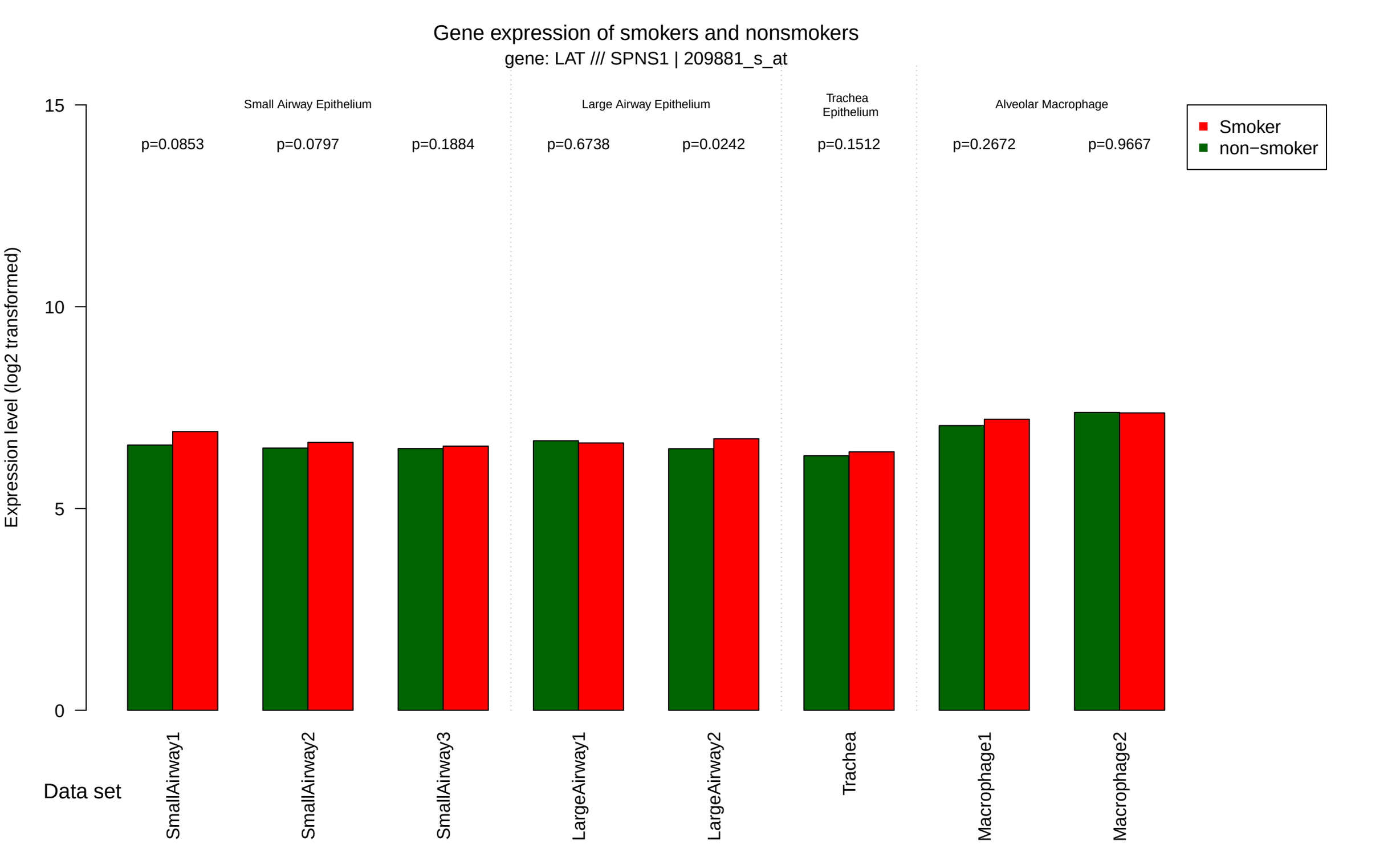
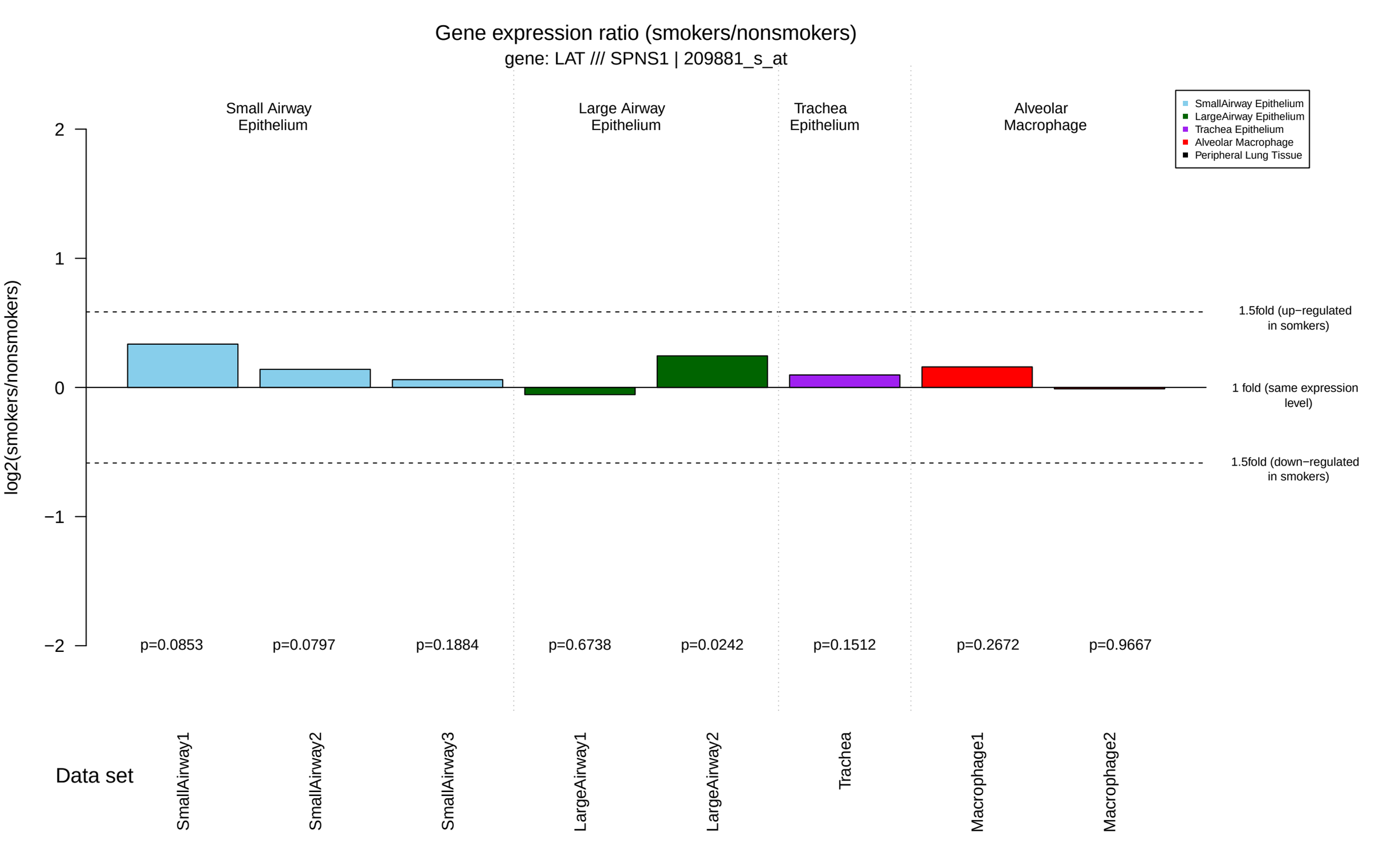
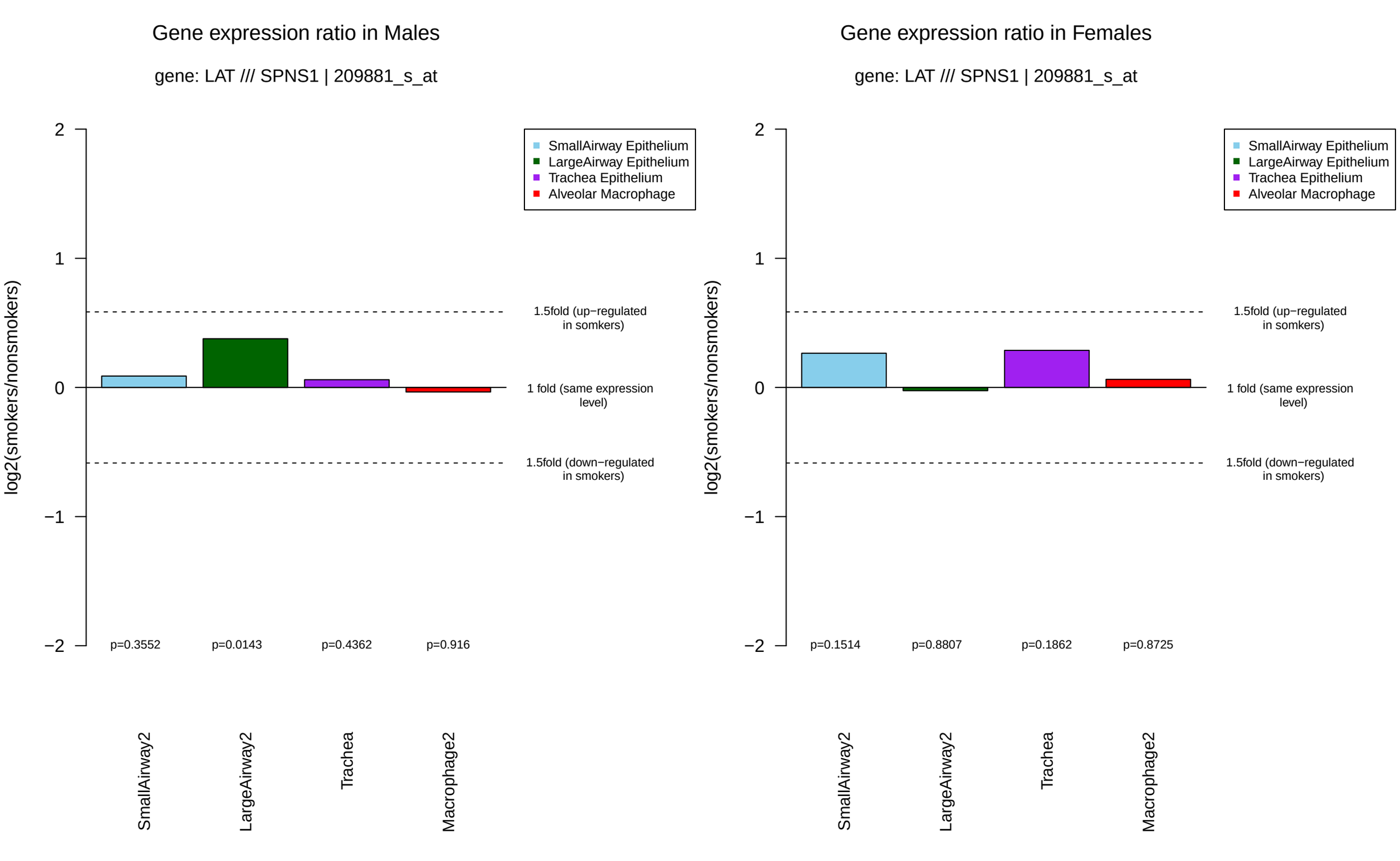
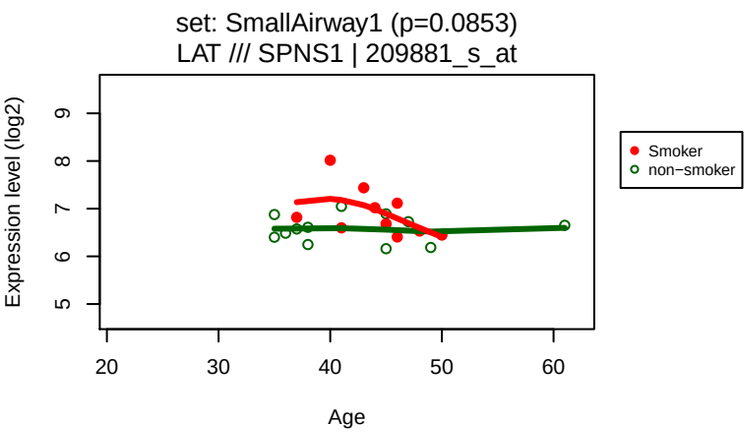
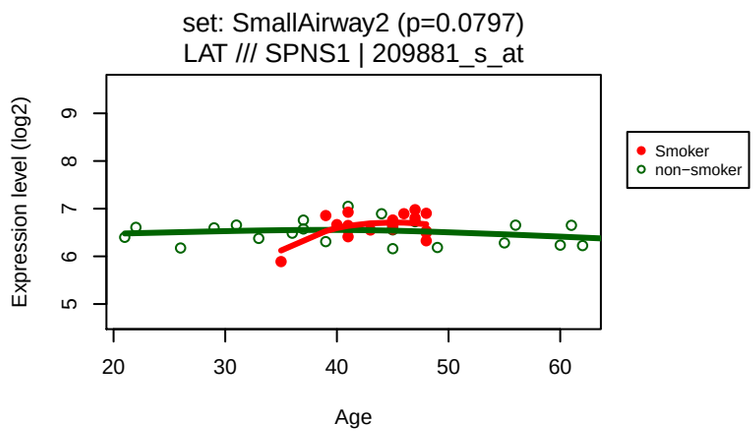
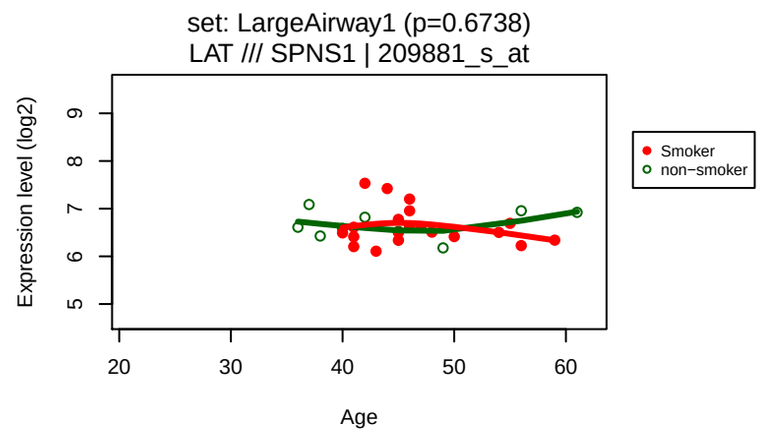

Gene expression of smokers and nonsmokers
gene: LAT /// SPNS1 | 209881_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0853
p=0.0797
p=0.1884
p=0.6738
p=0.0242
p=0.1512
p=0.2672
p=0.9667
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

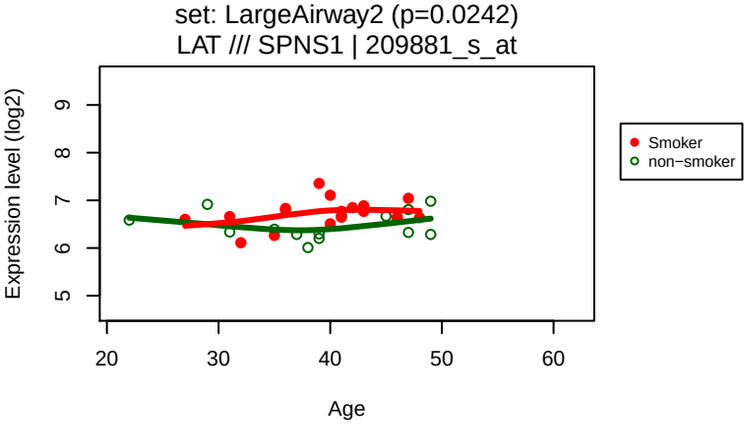

Gene expression ratio (smokers/nonsmokers)
gene: LAT /// SPNS1 | 209881_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0853
p=0.0797
p=0.1884
p=0.6738
p=0.0242
p=0.1512
p=0.2672
p=0.9667
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

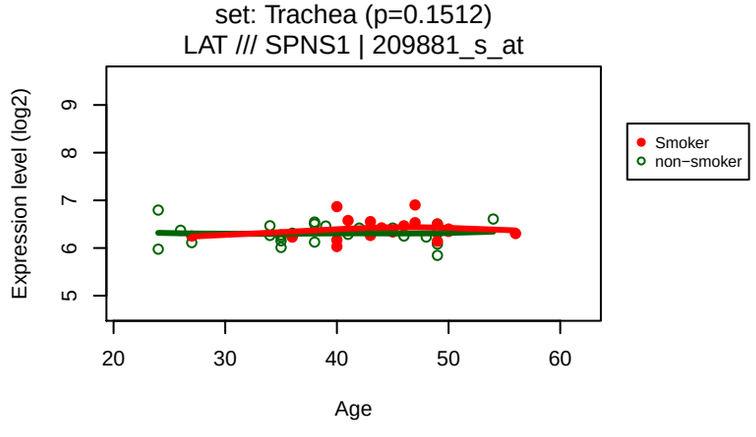

Gene expression ratio in Males
gene: LAT /// SPNS1 | 209881_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3552
p=0.0143
p=0.4362
p=0.916
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LAT /// SPNS1 | 209881_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1514
p=0.8807
p=0.1862
p=0.8725
SmallAirway2
LargeAirway2
Trachea
Macrophage2

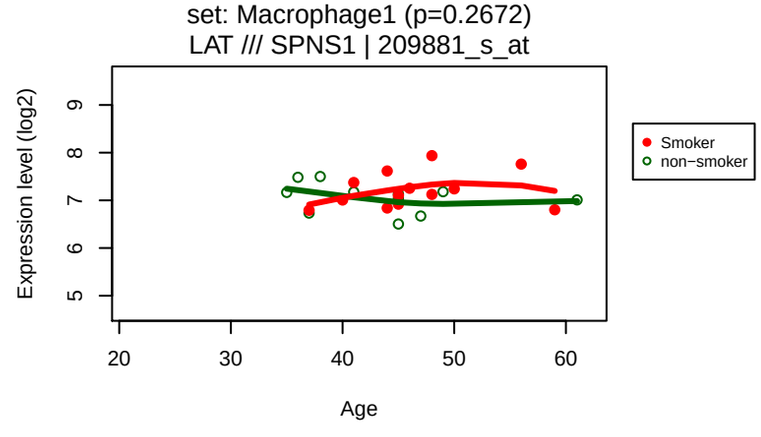

set: SmallAirway1 (p=0.0853)
LAT /// SPNS1 | 209881_s_at
Expression level (log2)
Age
Smoker
non−smoker

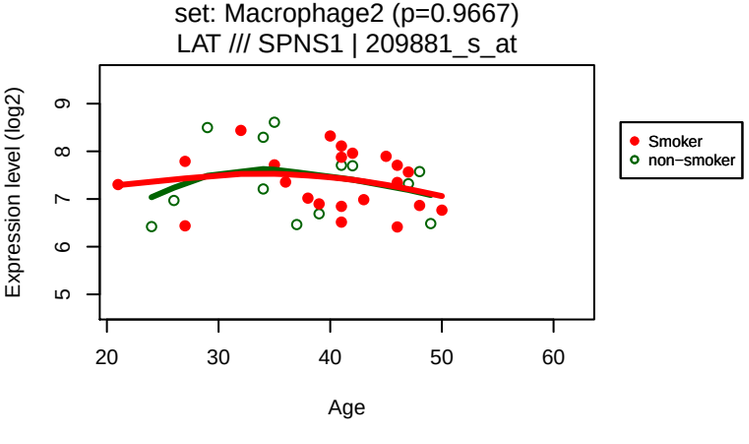

set: SmallAirway2 (p=0.0797)
LAT /// SPNS1 | 209881_s_at
Expression level (log2)
Age
Smoker
non−smoker

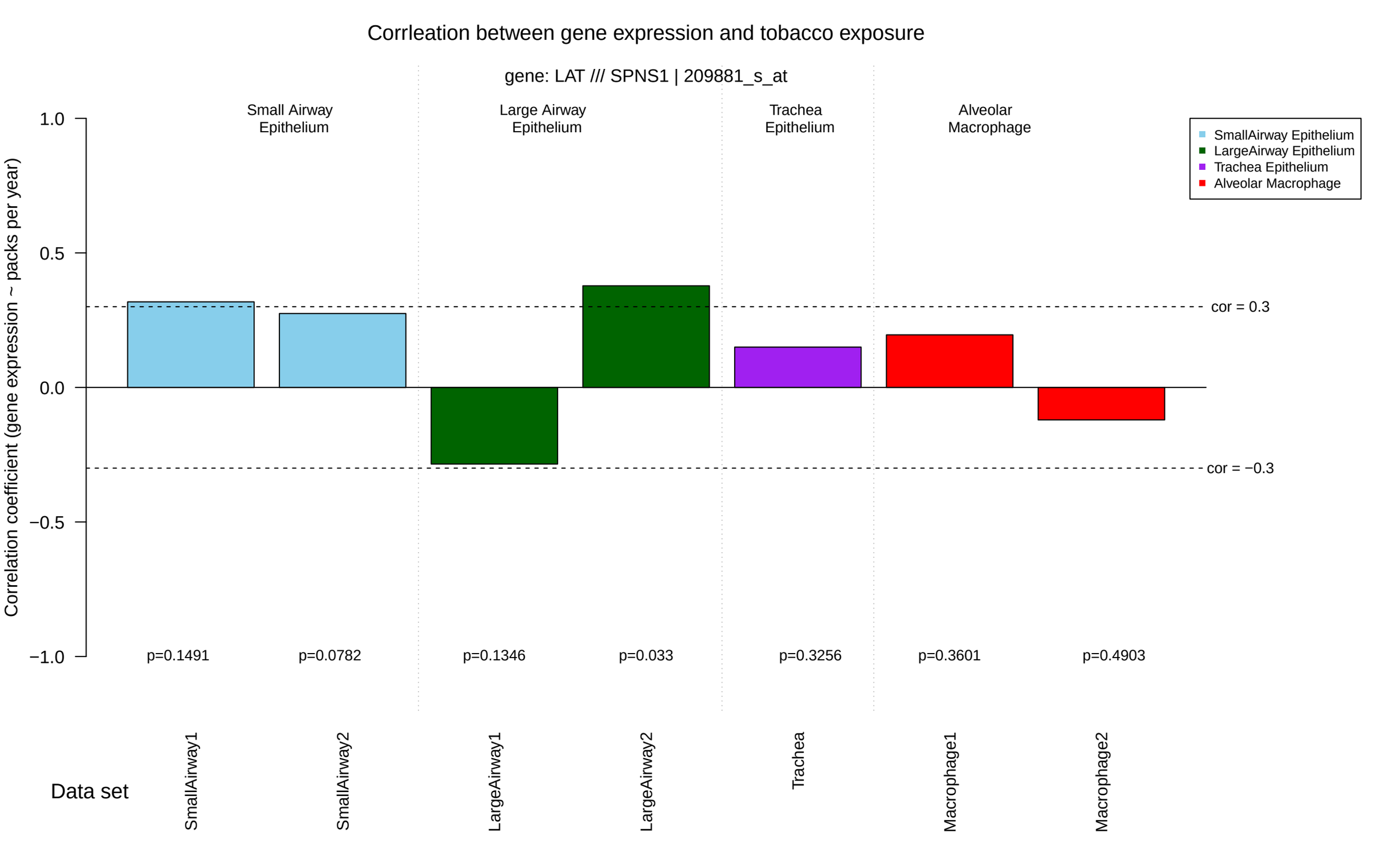

set: LargeAirway1 (p=0.6738)
LAT /// SPNS1 | 209881_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0242)
LAT /// SPNS1 | 209881_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1512)
LAT /// SPNS1 | 209881_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2672)
LAT /// SPNS1 | 209881_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9667)
LAT /// SPNS1 | 209881_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LAT /// SPNS1 | 209881_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1491
p=0.0782
p=0.1346
p=0.033
p=0.3256
p=0.3601
p=0.4903
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2